OSHE Framework Project

Project Description

Framework is an open source laptop and desktop brand focusing on the right to repair. Our goal is to build off this framework and develop ideas for the community based on what we feel is both wanted and applicable to the current modules. Keeping Framework's standard design constraints, which include the casing size for the expansion cards and the front panels slot sizes, we have created a variety of projects including an I2C/UART expansion card, an LCD front panel, an updated OLED Display, a Magnetic chagrin adapter, and a USB dongle hider expansion card. Designing these devices and outreach to the Framework community was the main goal of this project. This is the sunset semester for this project.

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I2C/UART Card

Overview and Features:

The I2C/UART card allows your computer to communicate over either of those protocols simply and effectively. The chip used is a MCP2221A chip in the SOIP14 form factor. The pinout of the switches are silkscreened onto the board so you can easily tell which mode the device is in. The 3d printed case, shown in the image collage below has cutouts of the case that are for the I/O, and the switch that allows you to change from I2C to UART. The bottom case has the pinout of the header on it in the form of a multi-color 3d print for ease of use. The PCB final design is shown in the collage below, fully assembled. The board also includes over voltage protection for the laptop, and the microchip present on the board. This was added to prevent burnouts of hardware, and to protect the USB ports of the framework utilizing the card. It is implemented using an Ideal Diode controller, combined with a MOSFET array chip to block anything greater than the supply voltage from backfeeding into the device. The array is tolerant to up to 24 volts of protection, which we saw as reasonable due to the intended use case of the device since it covers a wide range of potential supply voltages for devices that use I2C/UART for communication. As mentioned before, the MCP2221A chip that is on the board not only allows I2C but also UART. In order to use UART the 4 DIP switches need to be switched to a 1100 configuration. Once the card is in UART mode it can be plugged into any computer and used to communicate with microcontrollers or anything else that accepts UART. This functionality was tested with a raspberry pi pico 2 connected to an LED. A python script was used to send a UART signal to the Pico through the card. This signal told the pico to turn the LED on or off. Then the pico would send a UART signal back to the computer saying “LED on” or “LED off”. This tested both the RX and TX pins on the card.

Bill of Materials:

Part	Link	Amount	Cost ($)
I2C/UART PCBs (JLC)	N/A	5 (Minimum)	5.2
USB C Plugs Molex 1054440001 (Or equivalent)	Link	1	2.86
MCP2221A I2C/UART Bridge SOIP14	Link	1	2.53
Omron A6H-4101 DIP Switch	Link	1	2.50
LTC4412ES6 Ideal Diode Controller	Link	1	5.03
DMC2053UVT-13 P-N MOSFET Array	Link	1	0.57
5.1K 1206 SMD Resistor	Link	1	0.10
470K 1206 SMD Resistor	Link	10 (Cheaper than 1)	0.08
0.22uF SMD Capacitor	Link	1	0.18
0.47uF SMD Capacitor	Link	1	0.15
10uF 1206 SMD Capacitor	Link	1	0.15
2.54mm 1x4 Pin Pitch header	Link	1	0.45
Polyfuse 1.5A	Link	1	0.57
I2C/UART Card housing	N/A	1	N/A
M2x4 Screws	Link	2	5.69 (For 50)

Assembly Instructions:

1: Solder USB C Port to the pads labeled USB1, Use a fine tip soldering iron and a magnifying glass or microscope to inspect for bridging between contacts. Bridges may cause unwanted behavior or damage.

2: Line up the dot on the MCP2221A Package with the dot on the U1 Silkscreen, solder down a corner and then the rest of the pins. Kapton tape can be useful with this step. Inspect legs for bridges, as this can cause issues later on.

3: Solder the 0.47uF Capacitor to the C1 pads

4: Solder the 0.22uF Capacitor to the C2 pads

5: Solder the 10uF Capacitor to the C3 pads

6: Solder the fuse to the F1 pads

Optional feature assembly instructions

1: Solder the Omron switch to the SW1 pads

OR

1: Bridge switch override pads on backside of board

2: Solder the LTC4412xS6 chip to the U2 pads. Align the dot on the chip with the triangle on the board

3: Solder the DMC2053UVT chip to the Q1 pads. Align the dot on the chip with the triangle on the board

4: Solder the 470K Resistor to the R2 pads.

OR

2: Bridge OVP Bypass pads on backside of board

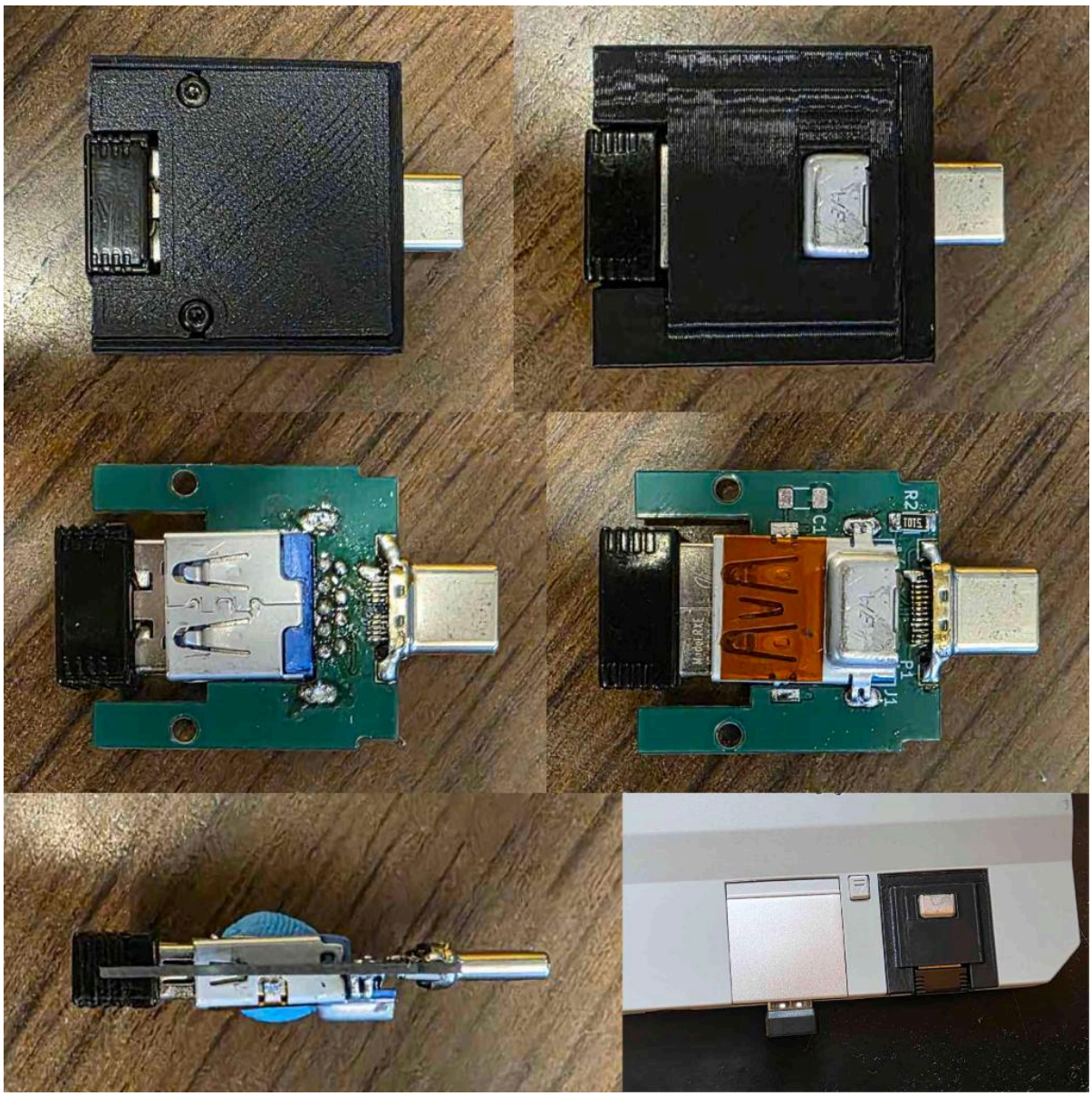
Usage:

Install MCP2221A drivers to your machine if not installed already.
Select either I2C or UART on the switch if you installed it. If not the card will always be I2C.
Connect peripherals in accordance with the pinout on the card itself.
Run script that can interface with the MCP2221A chip on the card (Usually python with EasyMCP2221 library)

Mouse Dongle Card

Overview and Features:

The main premise of this card was to provide a way for framework users to have the functionality of a wireless mouse, without the protruding dongle from the side of their machine. This was designed with modularity in mind, and ease of use and assembly relative to other options made by the community like the DongleHider+. We also wanted to have the full retention force that is created with a full size USB port and not one of those Type-C to USB-A adapters that fit inside male plugs. This card supports wireless dongles for mice, keyboards, wifi & bluetooth, and any other USB-A device that would fit inside the shell. The device and the board are both shown below, as well as in its intended use case.



Bill of Materials

Part	Link	Amount	Cost ($)
Mouse Dongle Card PCB	N/A	5 (Minimum)	5.2 (Estimate)
USB C Plugs Molex 1054440001 (Or equivalent)	Link	1	2.86
Inset USB-A Receptacle	Link	1	3.38
5.1K 1206 SMD Resistor	Link	1	0.10
Mouse Dongle Card housing	N/A	1	N/A
M2x4 Screws	Link	2	5.69 (For 50)

Assembly Instructions:

1: Solder the USB C port to the P1 pads on the board, be careful about bridging the pads

2: Place the USB A port into the 3d printed shell with the pins facing up. Make sure to put a usb plug into the port and bend the tabs on the sides inward to not interfere with the fit.

3: Place the pcb in the shell on top of the port. Make sure the silkscreen is facing down.

4: Line up the pin holes and mounting holes with corresponding parts on the port and solder them down.

5: Take the board out and solder the 5.1K resistor to the R2 pads.

6: Check for shorts between VBus and GND with a multimeter

7: Put the board back in the shell and secure the top cover with 2 M2x4 screws

Usage:

Plug in a wireless mouse dongle of your choice into the card and then plug the card into your machine

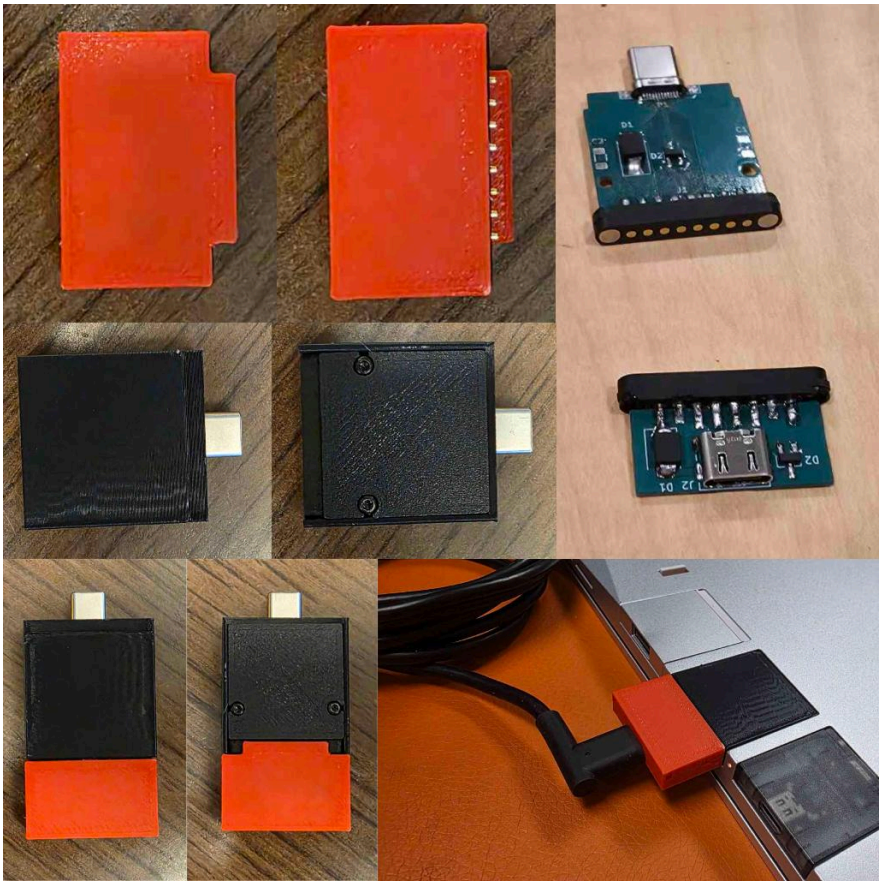
Magnetic Charger Card

Overview and Features:

The goal of this project was to create a magnetic charging card similar to Apple's MagSafe or Microsoft's Surface Connect. During the project, we successfully developed a card that can slot into all Framework laptops, with an adapter for USB Type-C cables. We extensively tested the charging capacity at 20 volts. While voltages above this were not thoroughly tested, the system should theoretically support higher levels. It was briefly tested at 48 volts; however, due to hardware constraints, it could not be evaluated for extended periods.

Each component used in both the charging card and adapter was carefully selected with consideration for cost and long-term availability. The most critical component was the magnetic connector. We selected an eight-pin pogo connector to provide ground redundancy in case the connector is attached in reverse, as well as to enable potential data transfer. For applications where data transfer is not required, and a smaller connector is preferred, a six-pin alternative could be used.

The enclosure for the expansion card was designed using the specifications provided in the official Framework documentation. A key challenge in this process was modifying the magnetic connector by reducing its edges to fit within the size constraints of the expansion card. For the adapter enclosure, an inlet was designed to accommodate the edges of the magnetic connector. During assembly and testing, sagging was observed in the structure, which was mitigated by adding an inlet and overhang, as shown in the figure below.



Bill of Materials:

Part	Link	Amount	Cost ($)
Magnetic Charger Card PCB	N/A	5 (Minimum)	40 (Estimate)
Magnetic Charger Adapter PCB	N/A	5 (Minimum)	40 (Estimate)
USB C Plugs Molex 1054440001 (Or equivalent)	Link	1	2.86
51V TVS Diode	Link	2	0.30
D+ D- ESD Protection TVS Diode	Link	2	0.27
Magnetic Connector Pair	Link	1	9.99
USB-C Receptacle	Link	1	1.87
Magnetic Charger Card housing	N/A	1	N/A
Magnetic Charger Adapter housing	N/A	1	N/A
M2x4 Screws	Link	2	5.69 (For 50)

Assembly instructions:

1: Solder the USB C Plug to the P1 header on the card
2: Solder the USB C Receptacle to the J2 header on the cable adapter
3: Solder the magnetic connector to both cards. Solder the connector so that both cards are face-up when properly connected, the connectors are NOT reversible.
4: Solder the SMBJ51ACT-ND Diode to D1 pads on the card and adapter PCB
5: Solder the ESD862DBZR Diode to the D2 pads on the card and adapter PCB
6: Check for shorts between any of the pins on the connector.

Front Panel Display (LCD)

Overview and Features:

The framework desktop display is designed with as aesthetically pleasing a design as we could create within the limits of the tile system of the Framework Desktop front panel. The display clips onto the front like any other modular tile, and requires some internal assembly for the device to function properly. Due to limitations of the framework desktop motherboard, the display utilizes a USB hub that needs to be properly connected to the mainboard via the connector for the front expansion bays, and the port needs to then be hooked up to the hub so you do not lose a port on your desktop. The display shows utilization for CPU, GPU, RAM, and VRAM for the AMD processor in the system.

Bill of Materials:

Part	Link	Amount	Cost ($)
LCD Display Housing	N/A	1	N/A
USB A-C Adapter	Link	1	7.39
USB Hub	Link	1	13.99
I2C LCD	Link	1	11.35
USB Type E to Type C Conversion Kit	Link	1	9.99
M3x10 Screws	Link	4	20 (for a kit)

Assembly instructions:

1: Desolder the included header on the LCD and solder female breadboard wires or other connectors to the holes.
2: Mount the LCD into the 3D printed housing with 4 M3x10 screws
3: Mount the USB hub of choice inside the desktop
4: Connect the Type-C End of the hub to the motherboard with the Type-E adapter.
5: Add other Type-E adapter to the hub.
6: Upload the I2C_LCD Code to the Microcontroller.

If this doesn't work, make sure to install the RP2040 board library, Adafruit TinyUSB, and LiquidCrystal_I2C

7: Connect the pins as follows

GND > GND, VDD > 5V, SDA > 0, SCL > 1

8: Mount microcontroller in safe location inside case and connect to hub with short USB cable

Front Panel Display (OLED)

Bill of Materials:

Part	Link	Amount	Cost
OLED Display Housing	N/A	1	N/A
OLED 128x64 Display	Link	1	13.59
RP2350-Zero Mini Development Board	Link	1	5.79
USB Hub	Link	1	13.99
USB Type E to Type C Conversion Kit	Link	1	9.99
M2.5 x 8 Screws	Link	4	20 (for a kit)

Assembly instructions:

1: Desolder the included header on the OLED and solder female breadboard wires or other connectors to the pads.
2: Mount the display into the 3D printed housing with 4 2.5 x 8 screws
3: Mount the USB hub of choice inside the desktop
4: Connect the Type-C End of the hub to the motherboard with the Type-E adapter.
5: Add other Type-E adapter to the hub.
6: Upload the SPI_OLED Code to the Microcontroller.

If this doesn't work, make sure to install the RP2040 board library, Adafruit TinyUSB, and U8g2lib

7: Connect the pins as follows.

GND > GND, VCC > 5V, SCK > 6, SDA > 3, RES > 8, DC > 7, CS > 5

8: Mount microcontroller in safe location inside case and connect to hub with short USB cable

Software Installation instructions (Windows):

1. Download the .EXE from the dist folder programs
2. If you would like to add it as a startup program press Win+R and enter "shell:startup" and click ok. This should open a folder named Startup
3. Copy the .exe into this folder

You may also need to rename your microcontroller for the software to recognize it:

1. Plug in the microcontroller to whichever device you will be using
2. Open device manager, find the microcontroller, and open its property window
3. Select the details tab and in the property drop down select device instance path
4. You should see a long string of numbers and letters in the "Value" field. Copy this string to your clipboard.
5. Open Notepad or any other text editor, and copy the below into it adding in your string in the specified brackets:
   Windows Registry Editor Version 5.00
   [THE STRING YOU COPIED HERE]
   "FriendlyName"="OSHE Framework Desktop Display"
6. Save the file with a .reg extension, such as "USBDeviceName.reg" ensuring you select "All Files" as the file type.
7. Right-click on the file you just saved and select "Merge" from the context menu. You may have to click more options to see this.
8. You will receive a prompt asking if you want to allow the registry editor to make changes to your computer. Click "Yes" to continue.
9. You should now see the updated name for the USB device in the Device Manager.

Software Installation instructions (Linux):

1. Download the latest release of the desktop display package.
2. Unzip it into a known location.
3. Navigate to the location of the wrapperM script
4. run **sudo chown root users wrapperM**
5. run **sudo chmod 0710 wrapperM**
6. run **sudo chmod u+s wrapperM**
7. Add this to your .profile file or other auto exec system
   a. cd <Path to script>
   b. ./wrapperM &
   c. *Remember the &, otherwise it WILL block login until it crashes!!!!!*
8. The script should now run when you login